

MARION, OH, USA

WMO: 725208

Lat: 40.616N

Lon: 83.064W

Elev: 994

StdP: 14.18

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04855

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	6.0	-6.4	4.1	2.0	-1.5	5.3	7.0	26.6	33.0	24.8	32.3	6.8	270	0.531

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	89.7	73.2	87.5	72.5	84.6	70.7	76.2	85.4	74.7	83.2	73.3	81.2	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	128.0	80.9	72.1	123.3	79.7	70.6	116.9	78.1	40.2	85.8	38.8	83.3	37.4	81.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.8	20.5	18.5	-6.1	92.2	7.2	2.8	-11.3	94.2	-15.5	95.9	-19.5	97.5	-24.7	99.5		
			-6.8	78.7	6.9	2.0	-11.7	80.2	-15.8	81.4	-19.6	82.5	-24.6	84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	26.2	29.2	38.6	50.8	61.7	70.0	73.1	71.2	65.0	53.4	41.9	31.6
	DBStd	18.85	12.34	11.60	11.08	9.42	8.39	6.21	5.31	5.00	7.25	9.12	9.56	10.50
	HDD50	2744	740	588	385	102	8	0	0	0	1	68	278	574
	HDD65	5849	1203	1002	818	433	165	25	4	8	88	375	694	1034
	CDD50	3177	2	6	33	126	372	601	715	657	451	174	35	5
	CDD65	806	0	0	1	7	64	176	254	200	88	16	0	0
	CDH74	6860	0	0	9	93	576	1462	2208	1599	787	124	2	0
	CDH80	2042	0	0	1	8	137	455	744	441	234	22	0	0
Wind	WSAvg	8.8	10.6	10.3	10.3	10.5	8.9	7.9	6.5	6.1	6.6	8.5	9.6	9.9
Precipitation	PrecAvg	37.3	2.9	2.0	2.6	3.6	4.1	4.4	4.2	3.7	3.2	2.8	2.8	2.8
	PrecMax	43.4	6.7	5.7	5.1	5.6	9.0	8.1	10.8	9.2	6.1	5.7	5.9	8.6
	PrecMin	25.4	1.1	0.7	1.2	1.6	1.7	1.3	1.5	1.4	1.1	0.6	0.7	0.9
	PrecStd	5.0	1.4	1.1	1.0	1.1	1.9	1.9	2.1	1.6	1.4	1.3	1.2	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.8	65.9	73.5	81.2	87.2	91.1	92.8	90.5	90.3	83.0	71.5	61.4
		MCWB	56.0	57.0	60.9	64.7	70.5	72.0	75.7	74.3	70.9	68.8	58.0	56.1
	2%	DB	54.8	57.2	66.6	76.8	83.8	88.2	89.8	88.0	86.4	77.2	64.9	55.8
		MCWB	51.7	52.1	57.0	61.2	67.4	71.7	74.7	73.4	70.5	64.3	56.7	52.2
	5%	DB	49.0	51.9	61.5	72.4	80.8	85.4	87.3	84.5	82.1	73.0	61.3	51.6
		MCWB	45.9	47.4	53.9	58.9	66.6	70.3	73.3	71.7	68.7	62.1	54.5	48.4
	10%	DB	43.5	45.5	56.6	67.7	76.8	82.1	84.2	82.0	78.8	68.7	57.0	46.4
		MCWB	39.7	40.9	50.2	56.3	64.6	68.9	71.6	70.7	66.2	60.5	51.4	43.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.3	58.1	62.4	67.7	72.8	75.5	78.5	77.9	75.1	71.3	62.4	56.9
		MCDB	59.0	63.9	71.3	77.2	82.0	85.7	88.2	86.0	85.0	80.4	67.5	60.3
	2%	WB	52.3	52.8	58.6	63.5	70.6	73.9	76.9	75.6	73.0	67.7	58.6	52.6
		MCDB	54.5	56.2	65.4	72.3	80.4	84.0	86.7	83.6	81.8	73.6	63.5	55.3
	5%	WB	46.4	47.1	55.0	60.6	68.4	72.5	75.2	74.1	71.0	64.3	55.3	48.3
		MCDB	48.5	51.5	60.4	69.2	77.1	82.0	83.7	81.4	78.5	70.3	59.8	50.7
	10%	WB	40.3	41.1	50.9	58.2	66.3	70.9	73.5	72.5	68.9	61.2	52.0	43.7
		MCDB	43.4	45.2	56.2	66.1	74.3	79.5	81.0	79.3	75.6	67.8	57.0	46.8
Mean Daily Temperature Range	5% DB	MDBR	14.6	15.3	17.5	21.2	20.2	19.1	19.2	19.8	21.2	19.4	16.7	13.0
		MCDBR	18.0	21.9	23.8	26.5	24.1	22.9	21.0	21.8	24.7	24.3	22.3	18.0
		MCWBR	16.2	17.5	15.8	15.0	11.9	9.7	8.8	10.1	11.4	13.2	15.8	16.1
	5% WB	MCDBR	17.2	20.4	21.1	22.6	19.8	19.0	18.7	18.3	20.2	20.0	19.1	17.3
		MCWBR	16.4	17.6	15.5	14.6	10.9	9.2	8.9	9.6	10.2	12.4	15.2	16.4
Clear-Sky Solar Irradiance	taub	0.306	0.328	0.354	0.397	0.431	0.448	0.454	0.443	0.389	0.360	0.324	0.303	
	taud	2.403	2.330	2.333	2.236	2.182	2.176	2.175	2.189	2.321	2.388	2.449	2.445	
	Ebn at Noon	270	279	284	278	270	264	261	261	269	265	261	262	
	Edh at Noon	26	32	36	43	46	47	46	44	36	30	25	23	
All-Sky Solar Radiation	RadAvg	495	781	1092	1476	1738	1909	1894	1713	1387	937	616	418	
	RadStd	46	96	104	129	145	142	112	75	117	79	84	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+110

Nomenclature: See separate page